Mayank Singh

Machine Learning ICT4 https://msingh27.github.io
Apple Inc. mayanksingh027@gmail.com

Education

Carnegie Mellon University, Robotics Institute
Master of Science in Robotics (GPA: 4.05/4.3) 2021 - 2022

Indian Institute of Technology Kharagpur 2012 - 2017
Bachelor of Science and Master of Science in Mathematics and Computing (GPA: 8.85/10.0)

Work Experience

Apple Inc., US
Machine Learning ICT4 Sept 2024 - Present
Machine Learning ICT3 Feb 2023 - Sept 2024

- Develop 3D generative models and computer-vision algorithms for object detection and tracking, pose estimation, and object-centric generative modeling.

Apple Inc., US
Technology Investigation Intern May 2022 – Aug 2022

- Worked on generative models for 3D computer vision.

Adobe Systems, India
Member of Technical Staff II Jan 2019 - Jan 2021
Member of Technical Staff I July 2017 - Jan 2019

- Developed a video tutorial recommendation system for Adobe Creative Cloud.
- Designed a reinforcement learning method to personalize the user experience.

Adobe Systems, India
Intern May 2016 – July 2016

- Developed a novel recommendation system by jointly modeling items and users.

Johns Hopkins University, US
Intern May 2015 - July 2015

- Worked on building a statically typed scripting language called BigBang.

Publications [Google Scholar]

- **Mayank Singh**, Michele Stoppa, Alvise Memo, Rui Yu, Harsha Kalli, Srimanth Gunturi, Muhammad Ahmed Riaz, Behrooz Shahsavari, Waleed Abdulla, David E. Jacobs. *Luce: Relightable Gaussians for 3D Asset Generation* . arXiv preprint 2026. [Paper].
  - Proposed a novel 3D representation that unifies geometry and PBR materials within a voxelized multi-modal Gaussian cloud.
  - Improved the state of the art in 3D generation from a single input image.
- Nikolaos Gkanatsios*, **Mayank Singh***, Zhaoyuan Fang, Shubham Tulsiani, Katerina Fragkiadaki. *Analogy-Forming Transformers for Few-Shot 3D Parsing*. ICLR 2023. [Paper].
  - Proposed a novel architecture named Analogical Networks that casts fine-grained 3D visual parsing as analogy-forming inference.
  - Improved the state of the art in 3D segmentation across many-shot, meta-learning, and few-shot settings.
- **Mayank Singh***, Nupur Kumari*, Puneet Mangla, Abhishek Sinha, Vineeth N Balasubramanian, Balaji Krishnamurthy. *Attributional Robustness Training using Input-Gradient Spatial Alignment*. ECCV 2020. [Paper].
- Puneet Mangla*, Nupur Kumari*, **Mayank Singh***, Balaji Krishnamurthy, Vineeth N Balasubramanian. *Data Instance Prior (DISP) in Generative Adversarial Networks*. WACV 2022. [Paper].
  - Proposed a transfer-learning method for GAN training that improved image-generation quality in low-data settings.

(* denotes equal contribution)

- Parth Patel*, Nupur Kumari*, **Mayank Singh***, Balaji Krishnamurthy. *LT-GAN: Self-Supervised GAN with Latent Transformation Detection.* WACV 2021. [Paper].
  - Proposed a self-supervised approach to improve the generation quality of images by estimating the GAN-induced transformation.
- Gunjan Aggarwal*, Abhishek Sinha*, Nupur Kumari*, **Mayank Singh***. *On the Benefits of Models with Perceptually-Aligned Gradients.* Towards Trustworthy ML, ICLR Workshop, 2020. [Paper].
- Nupur Kumari*, **Mayank Singh***, Abhishek Sinha*, Harshitha Machiraju, Balaji Krishnamurthy, Vineeth N Balasubramanian. *Harnessing the Vulnerability of Latent Layers in Adversarially Trained Models.* IJCAI 2019. [Paper].
- Puneet Mangla*, **Mayank Singh***, Abhishek Sinha*, Nupur Kumari*, Vineeth N Balasubramanian, Balaji Krishnamurthy. *Charting the Right Manifold: Manifold Mixup for Few-shot Learning.* WACV 2020. [Paper].
  - Proposed a novel few-shot image-classification method that leverages self-supervision and Manifold Mixup augmentation.
- **Mayank Singh***, Nupur Kumari*, Abhishek Sinha*, Balaji Krishnamurthy. *Understanding Adversarial Space through the lens of Attribution.* ECML PKDD Workshop 2018. [Paper].
- **Mayank Singh***, Abhishek Sinha*, Balaji Krishnamurthy. *Neural Networks in an Adversarial Setting and Ill-Conditioned Weight Space.* ECML PKDD Workshop 2018. [Paper]
- Tejus Gupta*, Abhishek Sinha*, Nupur Kumari*, **Mayank Singh***, Balaji Krishnamurthy *A Method for Computing Class-wise Universal Adversarial Perturbations.* arXiv preprint, 2019. [Paper]

(* denotes equal contribution)

US Patents

- **Mayank Singh**, Abhishek Sinha, Balaji Krishnamurthy. Systems and methods of training neural networks against adversarial attacks. **US11734565B2**
- **Mayank Singh**, Puneet Mangla, Nupur Kumari, Balaji Krishnamurthy, Abhishek Sinha. Classifying digital images in few-shot tasks based on neural networks trained using manifold mixup regularization and self-supervision. **US11308353B2**
- **Mayank Singh**, Balaji Krishnamurthy, Nupur Kumari, Puneet Mangla. Attributionally robust training for weakly supervised localization and segmentation. **US11875512B2**
- Pankhri Singhai, Sundeep Parsa, Piyush Gupta, Nupur Kumari, Nikaash Puri, **Mayank Singh**, Eshita Shah, Balaji Krishnamurthy, Akash Rupela. Machine-learning based multi-step engagement strategy modification. **US11816696B2**
- **Mayank Singh**, Nupur Kumari, Dhruv Khattar, Balaji Krishnamurthy, Abhishek Sinha. Generating trained neural networks with increased robustness against adversarial attacks. **US11829880B2**
- Pankhri Singhai, Sundeep Parsa, Piyush Gupta, Nikaash Puri, Eshita Shah, Balaji Krishnamurthy, Nupur Kumari, **Mayank Singh**, Akash Rupela. Machine-learning based multi-step engagement strategy generation and visualization. **US10609434B2**
- **Mayank Singh**, Parth Patel, Nupur Kumari, Balaji Krishnamurthy. Image generation and editing with latent transformation detection. **US11631156B2**

Academic Service

- **Peer Reviewer:** 2021 – Present
  - Completed 51 peer-review reports on 42 distinct conference and workshop submissions and journal manuscripts for ICML, ICLR, NeurIPS, WACV, IEEE TPAMI, IJCV, Pattern Recognition, IEEE Signal Processing Letters, Neural Networks, and workshops at ECCV and CVPR.
- **Research Assistant, Carnegie Mellon University** March 2021 – December 2022
  - Conducted research on 3D object detection, 3D reconstruction, and few-shot learning under the supervision of Prof. Katerina Fragkiadaki.

Languages and Libraries

Proficient: Python, PyTorch, NumPy, TensorFlow **Familiar:** C++, Java, Apache Spark, OpenMP, OpenCV